

3. Conditions and independence

Recall, Ω = sample space, $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ set of events

$P: \mathcal{F} \rightarrow \mathbb{R}$ is called probability (measure) if

- $P(\Omega) = 1$
- if $E_i \cap E_j = \emptyset \Rightarrow \sum P(E_i) = P(\bigcup E_i)$
- $P(E) \geq 0, \forall E \in \mathcal{F}$

If Ω is finite, then $\forall E \in \mathcal{F}, P(E) = \frac{|E|}{|\Omega|}$ is called uniform distribution.

I. Conditional probabilities

Example: Toss two dice, what is the probability that the sum of the two outcomes is 8?

$$|\Omega| = 36$$

$$E = \{\text{sum is } 8\} = \{(2,6), (3,5), (4,4), (5,3), (6,2)\} \Rightarrow P(E) = \frac{5}{36}$$

- What is the probability if we know that the first die is a 5?

$$\mathcal{F} = \{\text{first is } 5\} = \{(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)\}$$

$$P(\text{sum is } 8 \mid \mathcal{F}) = \frac{1}{|\mathcal{F}|} = \frac{1}{6}$$

it can happen only 1 way
due to our knowledge of the first roll
now the set of all possibilities is $\mathcal{F}$ (not Ω)

- What is the probability that the sum is 8 and the first roll is 5?

$$P(E \cap \mathcal{F}) = \frac{1}{36} \quad \text{Note that it's not the same, but} \quad \frac{P(E \cap \mathcal{F})}{P(\mathcal{F})} = \frac{1/36}{1/6} = \frac{1}{6}$$

def. Let $\mathcal{F} \in \mathcal{F}$ be an event with $P(\mathcal{F}) > 0$.

Then the conditional probability of $E \in \mathcal{F}$ with respect to $\mathcal{F}$ is

$$P(E|\mathcal{F}) = \frac{P(E \cap \mathcal{F})}{P(\mathcal{F})}$$

Prop. For $\mathcal{F} \in \mathcal{F}, P(\mathcal{F}) > 0$ the function $P(\cdot|\mathcal{F}): \mathcal{F} \rightarrow \mathbb{R}$ is a probability measure.

proof. There are 3 requirements a probability measure must satisfy.

$$1) P(\Omega|\mathcal{F}) = \frac{P(\Omega \cap \mathcal{F})}{P(\mathcal{F})} = \frac{P(\mathcal{F})}{P(\mathcal{F})} = 1$$

$$2) E_1, E_2, \dots \in \mathcal{F} \text{ are mutually exclusive} \Rightarrow P(\bigcup E_i|\mathcal{F}) = \frac{P((\bigcup E_i) \cap \mathcal{F})}{P(\mathcal{F})} = \frac{P(\bigcup (E_i \cap \mathcal{F}))}{P(\mathcal{F})} \stackrel{\text{me.}}{\leq} \sum \frac{P(E_i \cap \mathcal{F})}{P(\mathcal{F})} = \sum P(E_i|\mathcal{F})$$

$$3) E \in \mathcal{F} \Rightarrow P(E|\mathcal{F}) = \frac{P(E \cap \mathcal{F})}{P(\mathcal{F})} \geq 0$$

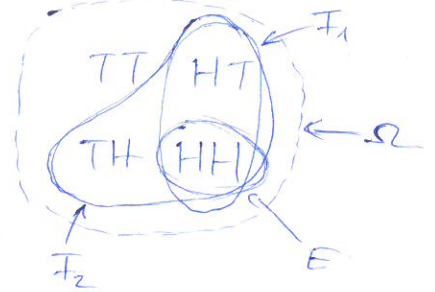
□

Example: Flip two coins!

$E = \{\text{both heads}\}$, $\mathcal{F}_1 = \{\text{first is heads}\}$, $\mathcal{F}_2 = \{\text{at least one is heads}\}$

$$P(E|\mathcal{F}_1) = P(E \cap \mathcal{F}_1) / P(\mathcal{F}_1) = P(E) / P(\mathcal{F}_1) = \frac{1/4}{1/2} = \frac{1}{2}$$

$$P(E|\mathcal{F}_2) = P(E \cap \mathcal{F}_2) / P(\mathcal{F}_2) = P(E) / P(\mathcal{F}_2) = \frac{1/4}{3/4} = \frac{1}{3}$$



Prop. Let $E_1, E_2, \dots \in \mathcal{F}$ such that $P(E_1 \cap \dots \cap E_{n-1}) > 0$, then

$$P(E_1 \cap \dots \cap E_n) = P(E_1)P(E_2|E_1)P(E_3|E_2 \cap E_1) \dots P(E_n|E_1 \cap \dots \cap E_{n-1})$$

proof. We substitute the definition of the conditional probability

$$\underbrace{P(E_1)}_{P(E_1)} \cdot \underbrace{\frac{P(E_2 \cap E_1)}{P(E_1)}}_{P(E_2|E_1)} \cdot \underbrace{\frac{P(E_3 \cap E_2 \cap E_1)}{P(E_2 \cap E_1)}}_{P(E_3|E_2 \cap E_1)} \dots \underbrace{\frac{P(E_n \cap E_{n-1} \cap \dots \cap E_1)}{P(E_{n-1} \cap \dots \cap E_1)}}_{P(E_n|E_{n-1} \cap \dots \cap E_1)} = P(E_1 \cap \dots \cap E_n) \quad \square$$

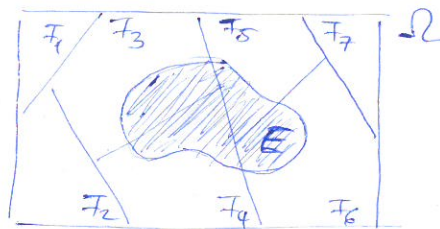
Thm. (Complete event systems)

Let $E \in \mathcal{F}$ and let $\{\mathcal{F}_i\}$ be a partition of Ω such that $\forall i: P(\mathcal{F}_i) > 0$.

$$\Rightarrow P(E) = \sum_i P(E|\mathcal{F}_i) \cdot P(\mathcal{F}_i)$$

proof. $P(E) = P(E \cap \Omega) = P(E \cap (\bigcup_i \mathcal{F}_i)) =$

$$= P(\bigcup_i (\mathcal{F}_i \cap E)) = \sum_i P(\mathcal{F}_i \cap E) = \sum_i \underbrace{\frac{P(E \cap \mathcal{F}_i)}{P(\mathcal{F}_i)}}_{P(E|\mathcal{F}_i)} P(\mathcal{F}_i) \quad \square$$



Example. (Insurance) In an insurance company they divide their customers into two groups:

- 1/ Good driver: 70% of customers causes an accident in a year with probability 0,2
- 2/ Bad driver: 30% of customers causes an accident in a year with probability 0,4

What is the probability, that a randomly chosen customer causes an accident?
 $A = \{\text{an accident}\}$, $G = \{\text{good driver}\}$, $B = \{\text{bad driver}\}$

$$P(A) = P(A|G)P(G) + P(A|B)P(B) = 0,2 \cdot 0,7 + 0,4 \cdot 0,3 = \underline{0,26}$$

Thm. (Bayes)

Let E be an event such that $P(E) > 0$.

Let $\{F_i\}$ be a complete system of events with $\forall i: P(F_i) > 0$.

Then

$$P(F_i | E) = \frac{P(E | F_i) P(F_i)}{\sum_j P(E | F_j) P(F_j)} \quad \text{holds for all } i.$$

proof:

$$P(F_i | E) = \frac{P(F_i \cap E)}{P(E)} = \frac{P(E | F_i) P(F_i)}{\sum_j P(E | F_j) P(F_j)} \quad \begin{array}{l} \leftarrow \text{definition of} \\ \text{conditional probability} \end{array}$$

$\leftarrow$ previous theorem

□

Example (Insurance again)

What is the probability, that an accident was caused by a good driver?

$$P(G | A) = \frac{P(A | G) P(G)}{P(A | G) P(G) + P(A | B) P(B)} = \frac{0.2 \cdot 0.7}{0.2 \cdot 0.7 + 0.4 \cdot 0.3} = \underline{0.54}$$

II. Independence

def. We say that the events $E, F \in \mathcal{F}$ are independent if

$$P(E \cap F) = P(E) P(F)$$

Note: For E, F independent events:

$$P(E | F) = \frac{P(E \cap F)}{P(F)} = \frac{P(E) P(F)}{P(F)} = P(E) \quad \leftarrow \begin{array}{l} \text{"knowing } F \text{" does not} \\ \text{change the probability} \\ \text{of } E \end{array}$$

Prop. If $E, F \in \mathcal{F}$ are independent, then E, F^c are also independent.

proof: $P(E \cap F^c) = P(E \setminus (E \cap F)) \underset{\text{prop.}}{=} P(E) - P(E \cap F) \underset{\text{indep.}}{=} P(E) - P(E)P(F) = P(E)(1 - P(F)) = P(E)P(F^c)$ □

def. Let $E_1, E_2, \dots$ be a countable sequence of events.

We say that $\{E_i\}$ are totally independent if

$$P(E_{i_1} \cap \dots \cap E_{i_k}) = P(E_{i_1}) P(E_{i_2}) \dots P(E_{i_k}), \quad \forall i_1, \dots, i_k \text{ such that } k \geq 1, i_k \neq i_\ell.$$

Example: Toss two dice! Let $E = \{\text{sum of outcomes is } 7\}$,

$F = \{\text{first roll is } 3\}$, $G = \{\text{second roll is } 4\}$,

$$E = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\} \Rightarrow P(E) = \frac{6}{36} = \frac{1}{6}, \quad P(F) = \frac{1}{6}, \quad P(G) = \frac{1}{6}$$

$$P(E \cap F) = P(E \cap G) = P(F \cap G) = P(\text{sum is } 7 \text{ \& first is } 3) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = P(E)P(F) = P(F)P(G) = P(E)P(G)$$

They are pairwise independent, but not totally independent!

$$P(E \cap F \cap G) = \frac{1}{36} \neq \frac{1}{6} \cdot \frac{1}{6} \cdot \frac{1}{6} = P(E)P(F)P(G)$$

Example. Let's make n many independent experiments.

The probability of the success of one experiment is p ($0 < p < 1$).

$E_i = \{i^{\text{th}} \text{ experiment is successful}\}$

a) $P(\text{all successful}) = ? = P(E_1 \cap \dots \cap E_n) \underset{\text{indep.}}{=} P(E_1) \cdot \dots \cdot P(E_n) = p^n$

b) $P(\text{at least one is successful}) = ? =$
 $= 1 - P(E_1^c \cap \dots \cap E_n^c) \underset{\substack{\uparrow \\ \text{complements are also indep. by prop.}}}{=} 1 - P(E_1^c)P(E_2^c) \dots P(E_n^c) = 1 - (1-p)^n$

c) $P(\text{exactly } k \text{ experiments were successful}) = ?$ ($0 \leq k \leq n$)
 What is the probability that the $i_1^{\text{th}}, \dots, i_k^{\text{th}}$ experiments succeeded, and the $j_1^{\text{th}}, \dots, j_{n-k}^{\text{th}}$ ones failed? ($\{i_1, \dots, i_k\} \cup \{j_1, \dots, j_{n-k}\} = \{1, \dots, n\}$)

$$P(E_{i_1} \cap \dots \cap E_{i_k} \cap E_{j_1}^c \cap \dots \cap E_{j_{n-k}}^c) = P(E_{i_1}) \cdot \dots \cdot P(E_{i_k}) P(E_{j_1}^c) \cdot \dots \cdot P(E_{j_{n-k}}^c) = p^k (1-p)^{n-k}$$

$$P(\text{exactly } k \text{ success}) = P\left(\bigcup_{1 \leq i_1 < \dots < i_k \leq n} \left\{ \begin{array}{l} \text{only the } i_1^{\text{th}}, \dots, i_k^{\text{th}} \\ \text{ones were successful} \end{array} \right\}\right) =$$

$$= \sum_{1 \leq i_1 < \dots < i_k \leq n} p^k (1-p)^{n-k} = \binom{n}{k} p^k (1-p)^{n-k}$$

Note: It is called binomial distribution.

III. Conditional independence

We have seen that $Q(\cdot) = P(\cdot|F)$ (if $P(F) > 0$) is a probability measure.

$$Q(G) > 0 \Rightarrow Q(E|G) = \frac{Q(E \cap G)}{Q(G)} = \frac{P(E \cap G|F)}{P(G|F)} = \frac{P(E \cap G \cap F)/P(F)}{P(G \cap F)/P(F)} = \frac{P(E \cap G \cap F)}{P(G \cap F)} = P(E|G \cap F)$$

def. We say that the events E and F are conditionally independent with respect to $G \in \mathcal{F}$ if

$$P(E \cap F|G) = P(E|G)P(F|G)$$

Note. In this case

$$P(E|F \cap G) = \frac{P(E \cap F \cap G)}{P(F \cap G)} = \frac{P(E \cap F \cap G)/P(G)}{P(F \cap G)/P(G)} = \frac{P(E \cap F|G)}{P(F|G)} \stackrel{\substack{\uparrow \\ \text{we've seen it}}}{=} \frac{P(E|G)P(F|G)}{P(F|G)} = P(E|G)$$

Example (Insurance again)

Suppose that the accidents in each year are independent from each other. A customer made an accident in 2016. What is the probability that he makes an accident in 2021?

$$P(A_{2021} | A_{2016}) \underset{\substack{\uparrow \\ \text{thm of complete event sys.} \\ \text{for the cond. prob. measure } P(\cdot | A_{2016})}}{=} P(A_{2021} | A_{2016} \cap G) P(G | A_{2016}) + P(A_{2021} | A_{2016} \cap B) P(B | A_{2016}) \underset{\substack{\uparrow \\ \text{conditional independence}}}{=} P(A_{2021} | G) P(G | A_{2016}) + P(A_{2021} | B) P(B | A_{2016})$$

0.54 ← we've seen it → 0.46

$= 0.2 \cdot 0.54 + 0.4 \cdot 0.46 = 0.292$